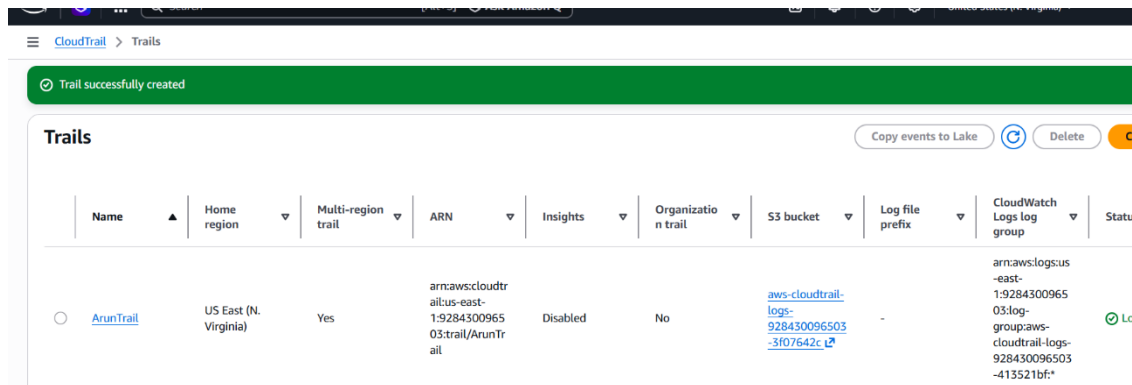


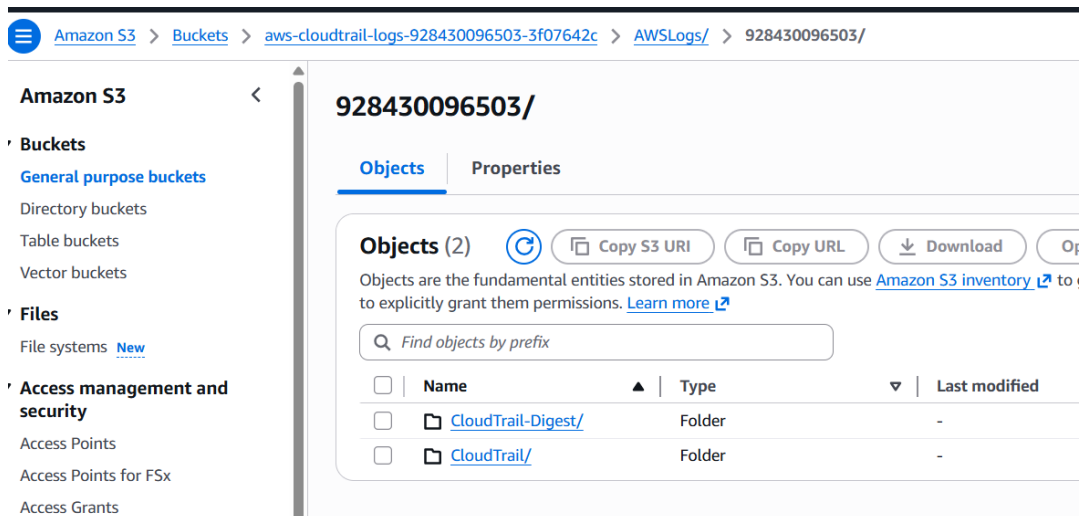
Task on CloudTrail & CloudWatch

1. Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

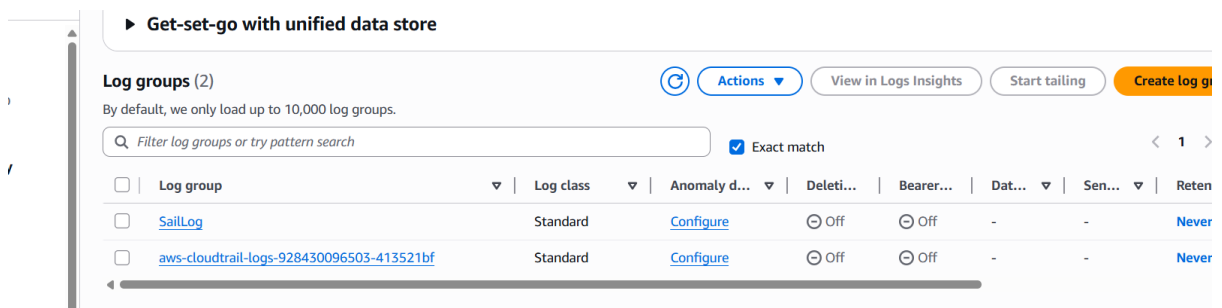
- CloudTrail Created

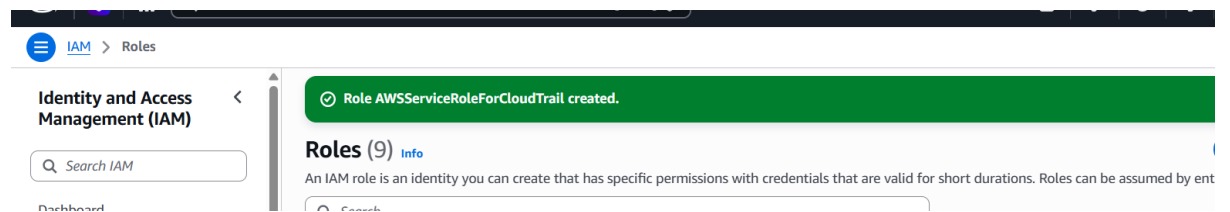
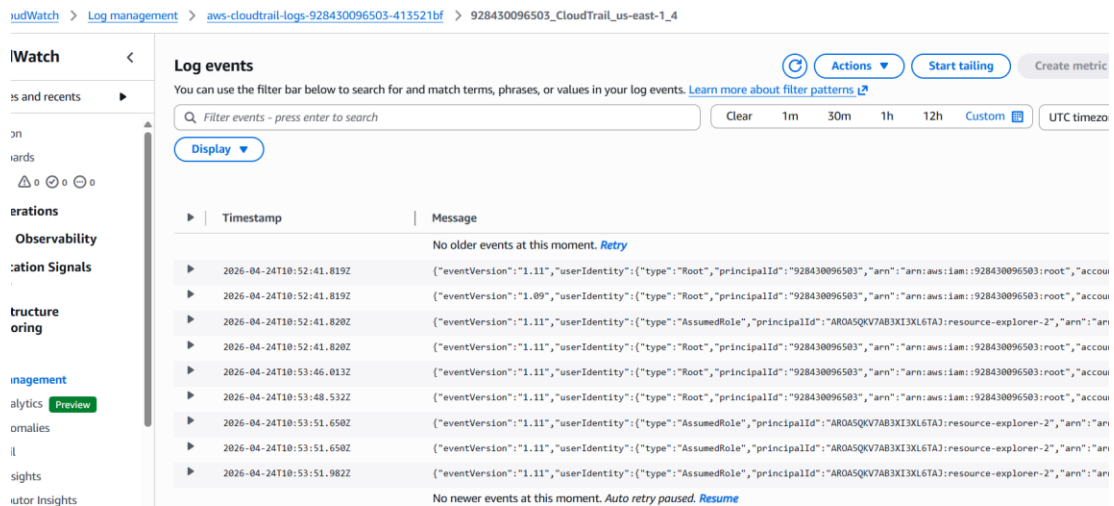


- Logs Stored in S3



- CloudWatch Logs

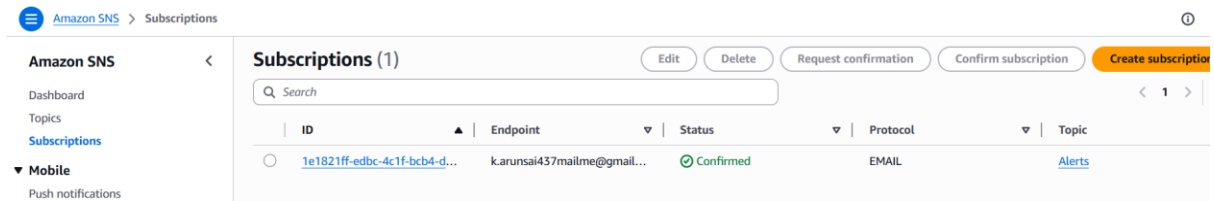




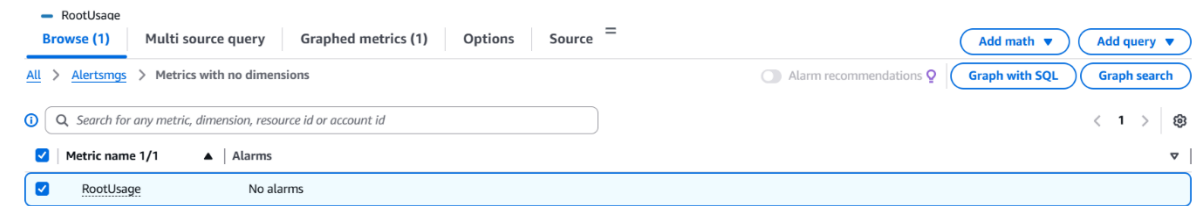
- **Created Alert in SNS**

- **Created Protocol Email in SNS**

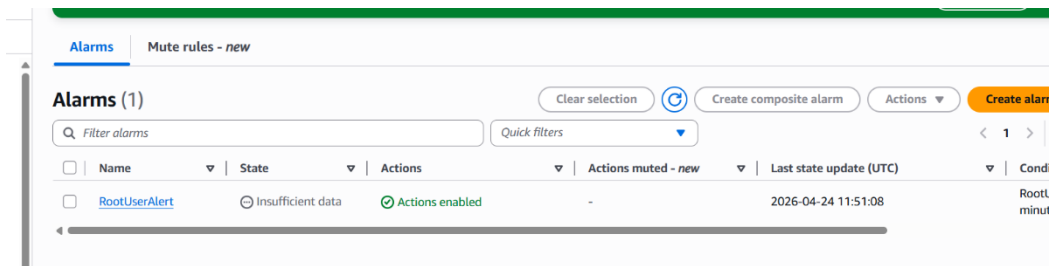
- Mail Subscription done



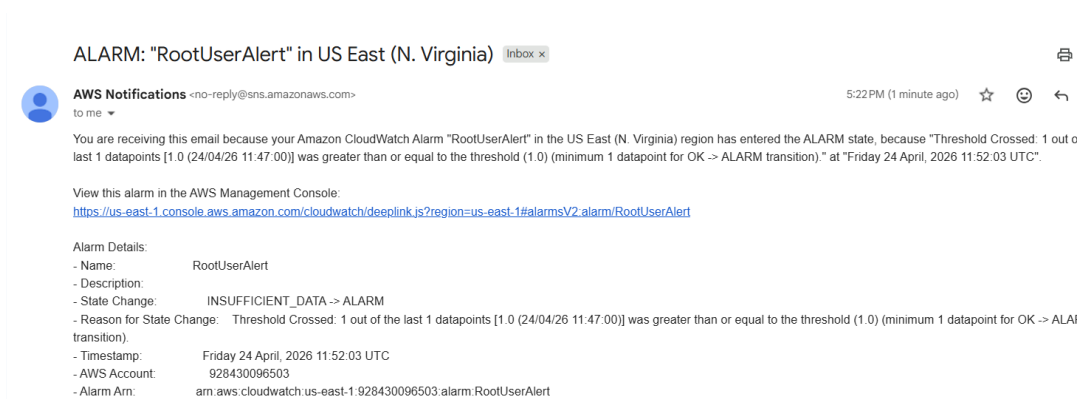
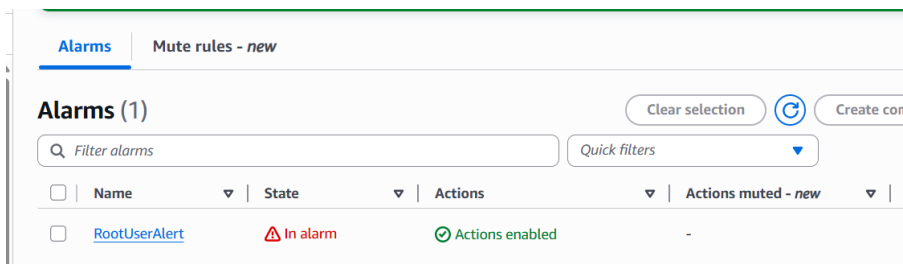
- Created Metric in CloudWatch Log Groups



- Created Alarm

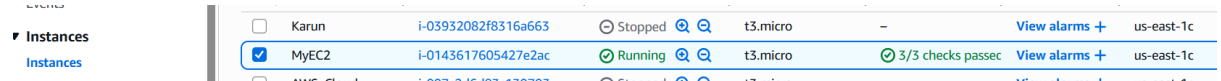


- Check in mail the alert alarm



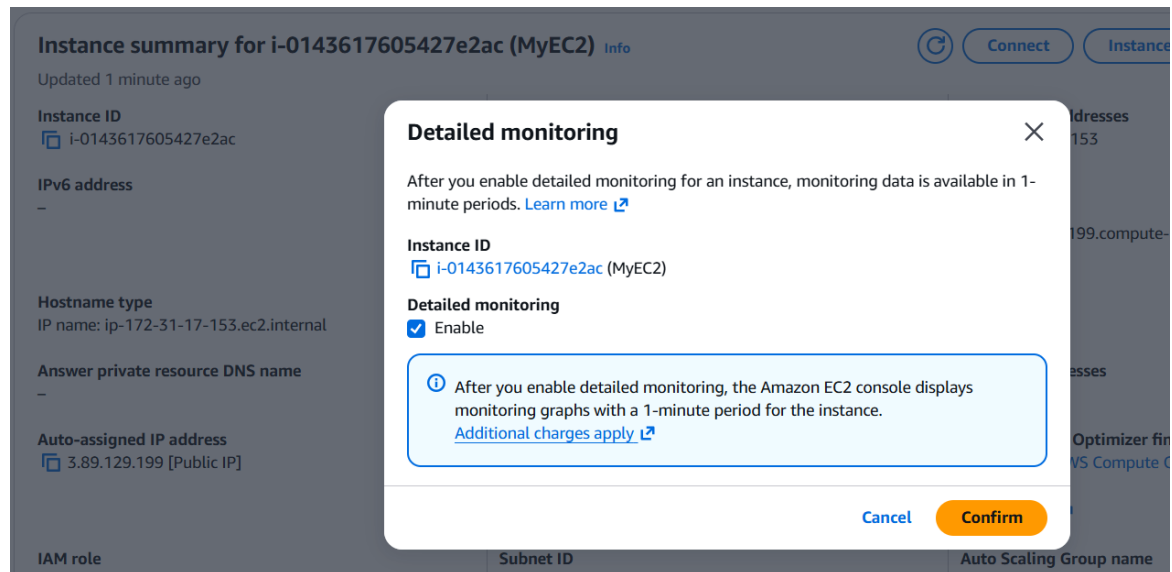
3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

- Run any instance default



☐	Karun	i-03932082f8316a663	Stopped	t3.micro	-	View alarms +	us-east-1c
☒	MyEC2	i-0143617605427e2ac	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c
☐	AWC Cloud	i-007e346487-1370702	Stopped	t3.micro	-	View alarms +	us-east-1c

- In that open Actions and enable the monitoring



Instance summary for i-0143617605427e2ac (MyEC2) Info

Updated 1 minute ago

Instance ID
i-0143617605427e2ac

IPv6 address
-

Hostname type
IP name: ip-172-31-17-153.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
3.89.129.199 [Public IP]

IAM role

Subnet ID

Auto Scaling Group name

Detailed monitoring

After you enable detailed monitoring for an instance, monitoring data is available in 1-minute periods. [Learn more](#)

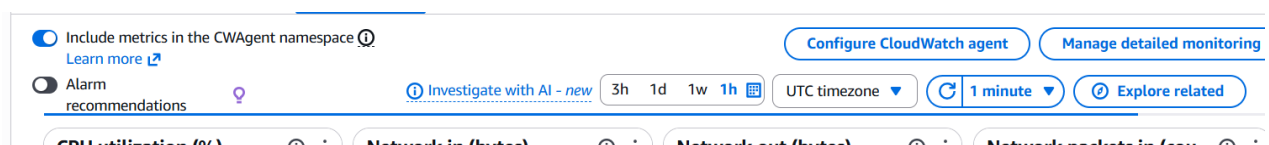
Instance ID
i-0143617605427e2ac (MyEC2)

Detailed monitoring
☒ Enable

After you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period for the instance. [Additional charges apply](#)

Cancel Confirm

- Change to 1 Minute for interval metrics



☒ Include metrics in the CWAgent namespace [Learn more](#)

☐ Alarm recommendations

[Investigate with AI - new](#)

3h 1d 1w 1h

UTC timezone

1 minute

[Explore related](#)

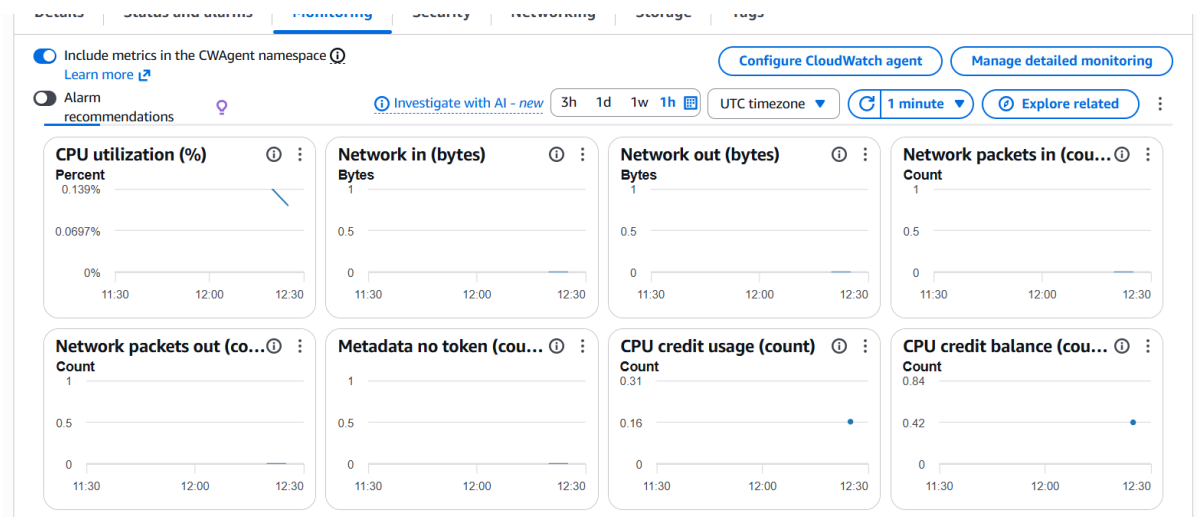
CPU utilization (%)

Network in (bytes)

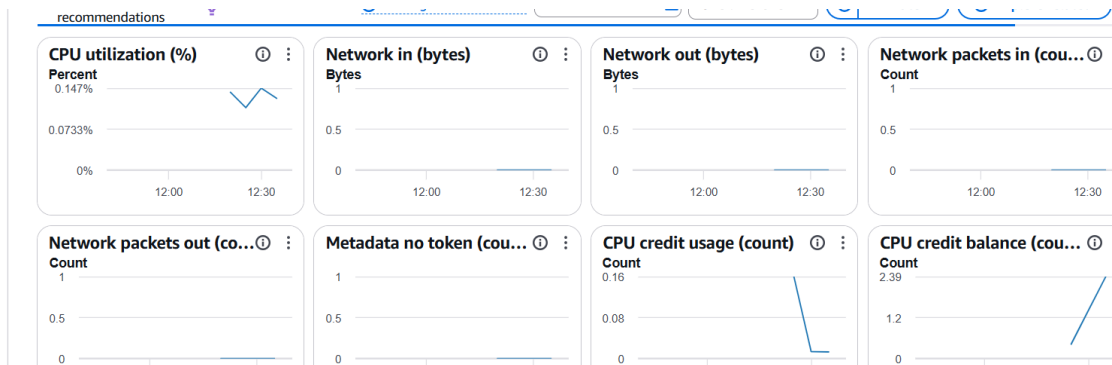
Network out (bytes)

Network packets in (count)

- Before it was zero



- **After Every 1 Minute the data is increasing Better accuracy for alarms**



4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

- **RUN any Ec2 in AWS and Connect with pem in gitbash**

instances	Choose filter set	Find instance by attribute or tag (case-sensitive)
instances		
Instance Types		
Launch Templates		
Not Requested		

- **Create an alert message in alarm the and set more than 70 CPU utilization**

Alarms (2)

Filter alarms

Quick filters

Clear selection

Create composite alarm

Actions

Create alarm

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
☐	HighCpuAlert	OK	Actions enabled	-	2026-04-24 13:52:10	CPUUtiliza minute
☐	RootUserAlert	In alarm	Actions enabled	-	2026-04-24 11:52:03	RootUsage minutes

- **Run the EC2 and use the command for increase CPU Utilization**

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ec2-user@ec2-23-22-51-92.compute-1.amazonaws.com
The authenticity of host 'ec2-23-22-51-92.compute-1.amazonaws.com' (23.22.51.92)
can't be established.
ED25519 key fingerprint is SHA256:+HE0rgPkPskx5NHCIloZXx2tekCRRxpWCQOjohEsRA.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:44: ec2-3-90-222-195.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-23-22-51-92.compute-1.amazonaws.com' (ED25519) t
o the list of known hosts.
```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
Last login: Fri Apr 24 15:14:49 2026 from 18.206.107.28
[ec2-user@ip-172-31-30-148 ~]$ yes > /dev/null
AC
[ec2-user@ip-172-31-30-148 ~]$ yes > /dev/null
AC
[ec2-user@ip-172-31-30-148 ~]$ $ yes > /dev/null
AC
[ec2-user@ip-172-31-30-148 ~]$ $ yes > /dev/null
AC
```

- Once the CPU utilization increase to 100 kill the process

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3225	ec2-user	20	0	221380	1028	932	R	52.4	0.1	2:57.04	yes
3226	ec2-user	20	0	221380	1016	924	R	52.4	0.1	2:54.14	yes
3227	ec2-user	20	0	221380	1024	932	R	52.4	0.1	2:52.21	yes
3228	ec2-user	20	0	221380	1024	932	R	52.4	0.1	2:50.56	yes
1	root	20	0	173084	17720	11116	S	0.0	1.9	0:01.08	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	slub_flushwq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	netns
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_hi
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthrea
14	root	20	0	0	0	0	S	0.0	0.0	0:00.05	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.08	rcu_preempt
16	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
18	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
20	root	rt	0	0	0	0	S	0.0	0.0	0:00.03	migration/1
21	root	20	0	0	0	0	S	0.0	0.0	0:00.06	ksoftirqd/1
23	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_hi
26	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	inet_frag_wq
28	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
29	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd
30	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
32	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	writeback
33	root	20	0	0	0	0	S	0.0	0.0	0:00.08	kcompactd0

- We can check in CloudWatch and also in mail



← ⓘ 🗑️ 📧 📁 ⋮

ALARM: "HighCpuAlert" in US East (N. Virginia) Inbox x

AWS Notifications <no-reply@sns.amazonaws.com>
to me ▼

You are receiving this email because your Amazon CloudWatch Alarm "HighCpuAlert" in the US East (N. Virginia) region has entered the ALARM state. The last 1 datapoints [99.99833336111064 (24/04/26 16:00:00)] was greater than the threshold (70.0) (minimum 1 datapoint for OK -> ALARM).

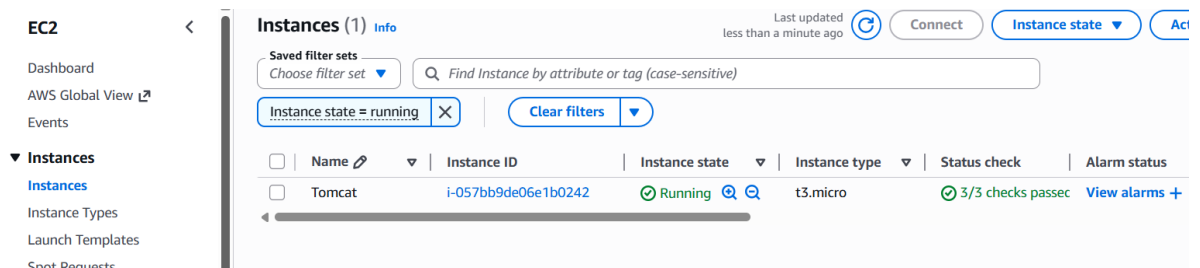
View this alarm in the AWS Management Console:
<https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/HighCpuAlert>

Alarm Details:

- Name: HighCpuAlert
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [99.99833336111064 (24/04/26 16:00:00)] was greater than the threshold (70.0)

5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

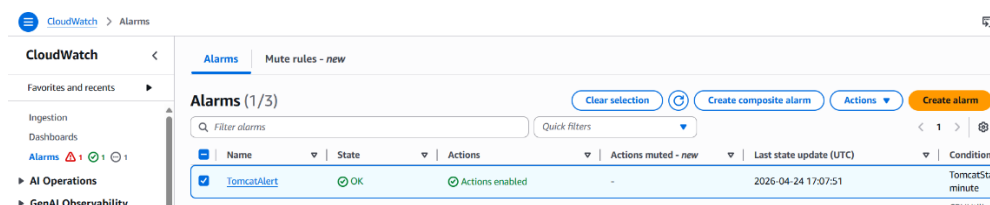
• New instance created and run



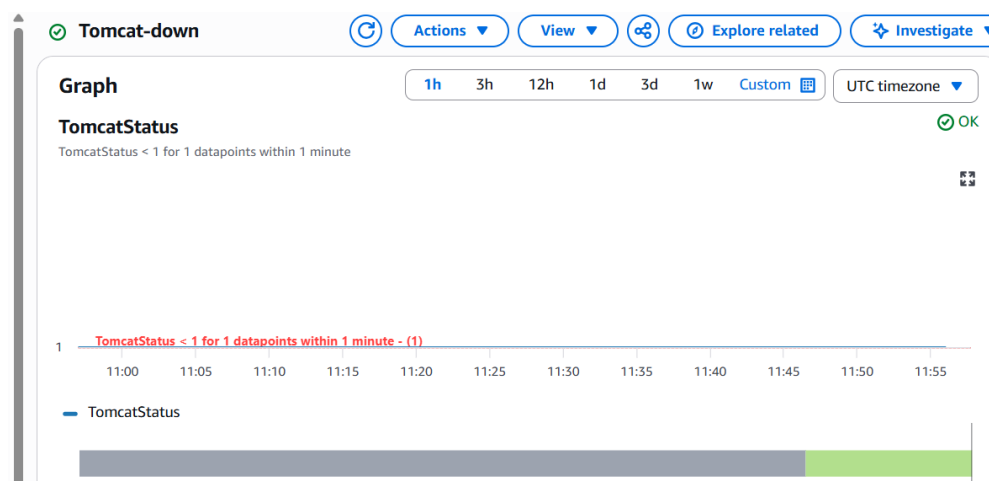
• Installed tomcat Apache and Run it

```
[ec2-user@ip-172-31-28-57 ~]$ ./tomcat-check.sh
[ec2-user@ip-172-31-28-57 ~]$ sudo su
[root@ip-172-31-28-57 ec2-user]# sudo pkill -f tomcat
[root@ip-172-31-28-57 ec2-user]# cd /opt/tomcat/bin
[root@ip-172-31-28-57 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
```

• Create an alarm for tomcat



• For alarm check in metric and value given low and attached to SNS



- Stop the tomcat server and check the alert alarm

```
[root@ip-172-31-28-57 opt]# mv script.sh tomcat-check.sh
[root@ip-172-31-28-57 opt]# ./tomcat-check.sh
[root@ip-172-31-28-57 opt]# sudo systemctl stop tomcat
[root@ip-172-31-28-57 opt]# ./tomcat-check.sh
[root@ip-172-31-28-57 opt]# sudo systemctl start tomcat
```

- We can see the alarm in Alarms in CloudWatch of tomcat

Alarms (1/3) Clear selection Refresh

Quick filters

☐	Name	State	Actions	Actions muted - new
☒	Tomcat-down	⚠ In alarm	✅ Actions enabled	-
☐	RootUserAlert	⚠ In alarm	✅ Actions enabled	-

- We can same thing in mail via SNS we get

1 of 80 <

ALARM: "Tomcat-down" in US East (N. Virginia) Inbox x

AWS Notifications <no-reply@sns.amazonaws.com> 5:29 PM (13 minutes ago) ☆ ☺ ↶

You are receiving this email because your Amazon CloudWatch Alarm "Tomcat-down" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of 1 last 1 datapoints [0.6666666666666666 (25/04/26 11:58:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Saturday 25 April, 2026 11:59:33 UTC".

View this alarm in the AWS Management Console:
<https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/Tomcat-down>

Alarm Details:

- Name: Tomcat-down
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [0.6666666666666666 (25/04/26 11:58:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Saturday 25 April, 2026 11:59:33 UTC
- AWS Account: 928430096503
- Alarm Arn: arn:aws:cloudwatch:us-east-1:928430096503:alarm:Tomcat-down

Threshold:

- The alarm is in the ALARM state when the metric is LessThanThreshold 1.0 for at least 1 of the last 1 period(s) of 60 seconds.

6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

- Created new instance Nginx in EC2 an run it

☐	Karun	i-05952082f8516a6b5	Stopped		t3.micro	-	View alarms +
☐	AWS	i-007afdc594e28ff2	Stopped		t3.micro	-	View alarms +
☒	Nginx	i-057bb9de06e1b0242	Running		t3.micro	3/3 checks pass	View alarms +

- Installed Nginx and run it

```
[ec2-user@ip-172-31-28-57 ~]$ systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-04-25 14:57:56 UTC; 10s ago
     Process: 3413 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 3414 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 3415 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 3416 (nginx)
      Tasks: 3 (limit: 1014)
     Memory: 3.4M
        CPU: 57ms
    CGroup: /system.slice/nginx.service
            └─3416 "nginx: master process /usr/sbin/nginx"
              └─3417 "nginx: worker process"
                └─3418 "nginx: worker process"

Apr 25 14:57:56 ip-172-31-28-57.ec2.internal systemd[1]: Starting nginx.service>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: the configurat>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: configuration>
```

- Created an alarm in cloud watch and threshold value is lower than 1

CloudWatch

Alarms Mute rules - new

Alarms (1/4)

Filter alarms Quick filters

Name	State	Actions	Actions muted - new	Last state update (UTC)
☒ Nginx-Alarm	OK	Actions enabled	-	2026-04-25 15:04:52

- Stop the nginx and check the alarm and mail

```
[ec2-user@ip-172-31-28-57 ~]$ sudo systemctl stop nginx
[ec2-user@ip-172-31-28-57 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: inactive (dead) since Sat 2026-04-25 15:05:52 UTC; 17s ago
     Duration: 7min 55.334s
    Main PID: 3416 (code=exited, status=0/SUCCESS)
        CPU: 59ms

Apr 25 14:57:56 ip-172-31-28-57.ec2.internal systemd[1]: Starting nginx.service>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: the configurat>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal nginx[3414]: nginx: configuration>
Apr 25 14:57:56 ip-172-31-28-57.ec2.internal systemd[1]: Started nginx.service>
Apr 25 15:05:52 ip-172-31-28-57.ec2.internal systemd[1]: Stopping nginx.service>
Apr 25 15:05:52 ip-172-31-28-57.ec2.internal systemd[1]: nginx.service: Deactiv>
Apr 25 15:05:52 ip-172-31-28-57.ec2.internal systemd[1]: Stopped nginx.service>
lines 1-14/14 (END)
```

- Check the alarm got the alarm on and mail also

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 2 1 1

AI Operations

GenAI Observability

Alarms

Mute rules - new

Alarms (1/4)

Filter alarms

Quick filters

	Name	State	Actions	Actions muted - new	Last state
☒	Nginx-Alarm	In alarm	Actions enabled	-	2026-04

- Got mail the alarm for Nginx

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10

ALARM: "Nginx-Alarm" in US East (N. Virginia)

Inbox x

AWS Notifications

<no-reply@sns.amazonaws.com>

to me

8:37 PM (0 minutes ago)

☆

You are receiving this email because your Amazon CloudWatch Alarm "Nginx-Alarm" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Cross last 1 datapoints [0.0 (25/04/26 15:06:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Saturday 25 April, 2026 15:07:52 UTC".

View this alarm in the AWS Management Console:
<https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/Nginx-Alarm>

Alarm Details:

- Name: Nginx-Alarm
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [0.0 (25/04/26 15:06:00)] was less than the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Saturday 25 April, 2026 15:07:52 UTC